

GATESVILLE, TX, USA

WMO: 722323

Lat: 31.421N

Lon: 97.797W

Elev: 276

StdP: 98.05

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 00362

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.1	-12.5	1.3	1.9	-9.9	1.7	3.5	10.5	13.5	9.2	14.6	1.4	10	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	38.0	22.9	37.4	23.1	36.2	23.1	25.5	31.7	25.1	31.7	24.7	31.1	3.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.6	26.5	23.7	19.1	26.4	22.9	18.3	26.3	79.3	32.2	77.9	31.2	76.1	30.4	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.6	6.9	DB	-7.3	40.4	2.7	1.7	-9.3	41.6	-10.8	42.6	-12.3	43.5	-14.3	44.8	(4)
			WB	-8.3	26.6	2.4	1.8	-10.0	27.9	-11.4	28.9	-12.8	29.9	-14.5	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	9.0	11.4	15.5	19.4	23.4	28.0	29.3	29.6	25.9	20.0	14.4	9.8
	DBStd	8.47	5.43	5.97	5.13	4.06	3.72	2.16	1.96	2.19	3.08	4.73	5.25	5.29
	HDD10.0	237	84	50	13	1	0	0	0	0	0	2	18	70
	HDD18.3	1098	293	203	115	35	8	0	0	0	1	38	137	268
	CDD10.0	3776	52	88	185	284	415	539	597	609	478	312	151	65
	CDD18.3	1595	3	8	28	68	164	289	339	351	229	91	20	4
	CDH23.3	18526	30	76	224	597	1548	3534	4354	4610	2497	876	147	33
	CDH26.7	9344	3	14	46	168	615	1855	2426	2647	1226	311	31	2
Wind	WSAvg	2.6	2.6	3.0	3.2	3.2	3.0	2.8	2.4	2.0	1.8	2.2	2.4	2.4
Precipitation	PrecAvg	845	51	68	80	67	104	103	54	62	61	85	76	55
	PrecMax	1277	115	181	268	151	249	240	200	226	123	155	277	167
	PrecMin	574	10	0	22	17	31	29	1	2	16	10	9	4
	PrecStd	180	30	49	55	42	60	56	48	61	33	48	67	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	28.0	30.1	32.8	35.9	38.1	39.1	40.1	37.9	33.7	29.9	26.7
		MCWB	14.9	16.2	17.8	20.6	22.1	23.0	22.5	22.6	21.8	21.1	20.9	17.9
	2%	DB	23.0	26.0	27.7	30.2	33.6	37.1	37.8	38.0	36.1	32.2	27.3	23.8
		MCWB	14.8	16.0	17.9	19.7	22.7	23.0	23.0	23.1	22.5	21.1	18.7	16.7
	5%	DB	21.2	22.9	26.2	28.0	32.2	35.9	37.2	37.5	34.0	30.1	24.8	21.4
		MCWB	15.1	15.6	17.7	19.5	22.5	23.1	23.2	23.3	22.4	20.8	17.8	15.8
	10%	DB	18.7	21.2	23.8	27.2	30.1	33.9	36.0	36.3	32.7	27.9	22.7	18.9
		MCWB	13.4	15.7	16.6	19.0	22.1	22.9	23.3	23.3	22.3	19.4	17.5	15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	20.2	21.7	23.5	26.3	25.7	25.8	25.8	25.5	24.5	22.4	20.6
		MCDB	21.9	23.8	25.8	28.3	28.3	32.2	31.9	32.1	31.5	28.7	25.8	23.0
	2%	WB	17.9	19.2	20.3	22.2	24.5	25.1	25.3	25.4	24.9	23.5	21.4	19.2
		MCDB	20.6	22.1	24.0	27.0	29.9	31.8	32.0	32.1	30.5	27.5	24.4	22.1
	5%	WB	16.2	18.0	19.5	21.5	23.8	24.6	24.9	25.0	24.3	22.7	20.1	17.6
		MCDB	19.0	20.7	23.5	26.1	28.9	31.0	31.5	31.9	29.5	26.7	23.1	19.5
	10%	WB	13.8	16.6	18.8	20.5	23.2	24.0	24.4	24.5	23.8	22.0	19.0	15.7
		MCDB	18.0	19.8	22.1	24.9	28.1	30.4	30.8	31.4	28.9	26.0	21.8	18.6
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.2	12.5	12.3	11.2	11.8	11.8	12.2	12.0	12.8	12.6	12.1
		MCDBR	16.1	15.0	14.0	13.6	13.1	13.6	13.5	13.6	13.6	13.7	13.8	15.3
		MCWBR	9.4	8.5	6.9	6.4	4.7	3.2	2.9	3.1	3.9	5.6	7.2	9.4
	5% WB	MCDBR	12.7	12.0	10.8	10.9	10.5	10.8	10.9	11.4	10.8	10.4	10.6	12.0
	MCWBR	9.3	8.4	6.4	6.2	4.9	3.4	3.1	3.1	3.4	5.5	7.3	9.4	
Clear-Sky Solar Irradiance	taub	0.312	0.325	0.352	0.392	0.423	0.424	0.436	0.427	0.396	0.367	0.339	0.317	
	taud	2.515	2.482	2.421	2.334	2.304	2.341	2.345	2.353	2.386	2.453	2.473	2.505	
	Ebn at Noon	919	936	930	900	869	864	852	857	873	878	877	892	
	Edh at Noon	86	98	112	127	132	127	126	123	115	100	90	83	
All-Sky Solar Radiation	RadAvg	3.02	3.55	4.59	5.53	5.85	6.66	6.77	6.26	5.23	4.18	3.17	2.67	
	RadStd	0.32	0.57	0.40	0.51	0.55	0.53	0.51	0.38	0.44	0.62	0.39	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+73

Nomenclature: See separate page